

27.2 Ridge regression

The oldest remedy for collinearity is also the simplest: when $\mathbf{X}^\top \mathbf{X}$ has eigenvalues close to zero, add a positive constant to its diagonal before inverting. Hoerl and Kennard (1970) proposed this in statistics; numerical analysts knew it as Tikhonov regularization. The constant bounds the variance at the price of bias towards the origin. Exercise (B1) and Exercise (C1) met the estimator briefly.

27.2.1 Centring, scaling and the intercept

A penalty should not touch the intercept, or shifting the response by a constant would change the fitted slopes. Leaving the intercept unpenalized is equivalent to centring, for every penalty.

Proposition 27.2.1. An unpenalized intercept

Let $\mathbf{X}_0$ be an $n \times p$ matrix of regressors with column means $\bar{\mathbf{x}}_0$, let $\mathbf{X} = \mathbf{X}_0 - \mathbf{1}\bar{\mathbf{x}}_0^\top$ and $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}$ be the centred regressors and response, and let $P : \mathbb{R}^p \rightarrow [0, \infty]$ be any function. Then $(\beta_0, \mathbf{b})$ minimizes

$$\|\mathbf{y} - \beta_0\mathbf{1} - \mathbf{X}_0\mathbf{b}\|^2 + P(\mathbf{b})$$

iff $\mathbf{b}$ minimizes $\|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2 + P(\mathbf{b})$ and $\beta_0 = \bar{y} - \bar{\mathbf{x}}_0^\top \mathbf{b}$.

Proof. For fixed $\mathbf{b}$, the first term is the residual sum of squares of regressing $\mathbf{y} - \mathbf{X}_0\mathbf{b}$ on $\mathbf{1}$, minimized uniquely by the mean $\beta_0 = \bar{y} - \bar{\mathbf{x}}_0^\top \mathbf{b}$, with minimum $\|(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)(\mathbf{y} - \mathbf{X}_0\mathbf{b})\|^2 = \|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2$. So the joint minimum is the minimum over $\mathbf{b}$ of $\|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2 + P(\mathbf{b})$, attained exactly at the pairs described. ■

From now on the regressors are centred and the response is centred, so $\mathbf{X}$ has no intercept column and $\mathbf{1}^\top \mathbf{X} = \mathbf{0}^\top$. A penalty also treats all coefficients as commensurable, so the regressors should be on a common scale (Exercise (C1)). In this chapter's examples every column has unit standard deviation, so $\mathbf{X}^\top \mathbf{X} = (n-1)\mathbf{R}$ with $\mathbf{R}$ the correlation matrix, and a coefficient is the change in mean response per standard deviation of its regressor.

27.2.2 The estimator

Definition 27.2.1. Ridge regression

For $\lambda \geq 0$, a **ridge estimate** $\hat{\boldsymbol{\beta}}_\lambda^R$ minimizes

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2. \quad (27.2.1)$$

The constant λ is the **ridge parameter** (or tuning parameter). The **ridge trace** is the curve $\lambda \mapsto \hat{\boldsymbol{\beta}}_\lambda^R$.

For $\lambda = 0$ this is least squares. For $\lambda > 0$ it is uniquely defined whatever the rank of $\mathbf{X}$, and its

behaviour is transparent in the coordinates of Eq. 27.1.1. Let $\mathbf{X} = \sum_{j \leq r} d_j \mathbf{u}_j \mathbf{v}_j^\top$ be the SVD of $\mathbf{X}$, of rank r , and $c_j = \mathbf{u}_j^\top \mathbf{y}$.

Theorem 27.2.2. Ridge regression

Let $\lambda > 0$.

- (a) (*Closed form.*) The criterion Eq. 27.2.1 has the unique minimizer

$$\hat{\boldsymbol{\beta}}_\lambda^R = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} = \sum_{j \leq r} \frac{d_j}{d_j^2 + \lambda} c_j \mathbf{v}_j.$$

If $r = p$, then $\hat{\boldsymbol{\beta}}_\lambda^R = \sum_j f_j(\lambda) \hat{\alpha}_j \mathbf{v}_j$ with **shrinkage factors** $f_j(\lambda) = d_j^2 / (d_j^2 + \lambda) \in (0, 1)$, decreasing in λ and smallest for the smallest d_j .

- (b) (*Fitted values.*) $\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda^R = \mathbf{H}_\lambda \mathbf{y}$ with $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top = \sum_{j \leq r} f_j(\lambda) \mathbf{u}_j \mathbf{u}_j^\top$. The **effective degrees of freedom** $\text{df}(\lambda) = \text{tr } \mathbf{H}_\lambda = \sum_{j \leq r} f_j(\lambda)$ decrease strictly from r (as $\lambda \rightarrow 0$) to 0 (as $\lambda \rightarrow \infty$).

- (c) (*Moments.*) If $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ and $r = p$, then

$$E \hat{\boldsymbol{\beta}}_\lambda^R - \boldsymbol{\beta} = -\lambda (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \boldsymbol{\beta},$$

$$\text{Cov}(\hat{\boldsymbol{\beta}}_\lambda^R) = \sigma^2 \sum_j \frac{d_j^2}{(d_j^2 + \lambda)^2} \mathbf{v}_j \mathbf{v}_j^\top,$$

and, with $\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}$, the total mean squared error is

$$E \|\hat{\boldsymbol{\beta}}_\lambda^R - \boldsymbol{\beta}\|^2 = \sigma^2 \sum_j \frac{d_j^2}{(d_j^2 + \lambda)^2} + \lambda^2 \sum_j \frac{\alpha_j^2}{(d_j^2 + \lambda)^2}. \quad (27.2.2)$$

- (d) (*The path.*) Unless $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$, $\|\hat{\boldsymbol{\beta}}_\lambda^R\|$ decreases strictly in λ , with limit 0 as $\lambda \rightarrow \infty$; and $\hat{\boldsymbol{\beta}}_\lambda^R \rightarrow \mathbf{X}^+ \mathbf{y}$, the least squares estimate of smallest norm, as $\lambda \rightarrow 0$.
- (e) (*Bayesian reading.*) If $\mathbf{Y} \mid \boldsymbol{\beta} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$ with σ^2 known and $\boldsymbol{\beta} \sim N_p(\mathbf{0}, (\sigma^2/\lambda) \mathbf{I})$, then $\boldsymbol{\beta} \mid \mathbf{Y} \sim N_p(\hat{\boldsymbol{\beta}}_\lambda^R, \sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1})$. With σ^2 unknown and the normal-inverse-gamma prior of Theorem 7.6.2 with $\mathbf{m}_0 = \mathbf{0}$ and $\mathbf{V}_0 = \lambda^{-1} \mathbf{I}$, the posterior mean of $\boldsymbol{\beta}$ is again $\hat{\boldsymbol{\beta}}_\lambda^R$.

Proof.

- (a) Uniqueness and the first formula are Exercise (A2): $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is positive definite, and Proposition 1.11.2 applies. Complete $\mathbf{v}_1, \dots, \mathbf{v}_r$ to an orthonormal basis $\mathbf{v}_1, \dots, \mathbf{v}_p$ of $\mathbb{R}^p$. Then $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I} = \sum_{j \leq r} (d_j^2 + \lambda) \mathbf{v}_j \mathbf{v}_j^\top + \sum_{j > r} \lambda \mathbf{v}_j \mathbf{v}_j^\top$, whose inverse has the same eigenvectors and reciprocal eigenvalues (Theorem 1.7.2). Since $\mathbf{X}^\top \mathbf{y} = \sum_{j \leq r} d_j c_j \mathbf{v}_j$, the second formula follows. When $r = p$, $d_j c_j / (d_j^2 + \lambda) = f_j(\lambda) c_j / d_j = f_j(\lambda) \hat{\alpha}_j$.
- (b) $\mathbf{X} \mathbf{v}_j = d_j \mathbf{u}_j$, so $\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda^R = \sum_{j \leq r} f_j(\lambda) c_j \mathbf{u}_j = \sum_j f_j(\lambda) \mathbf{u}_j \mathbf{u}_j^\top \mathbf{y}$. The trace of $\mathbf{u}_j \mathbf{u}_j^\top$ is one. Each f_j decreases strictly from 1 to 0.
- (c) These are the formulas of Exercise (B1) in the present notation (its d_j and $\boldsymbol{\gamma}$ are our d_j^2 and $\boldsymbol{\alpha}$); they are also Proposition 27.1.2(a) with the factors $f_j(\lambda)$.
- (d) $\|\hat{\boldsymbol{\beta}}_\lambda^R\|^2 = \sum_{j \leq r} d_j^2 c_j^2 / (d_j^2 + \lambda)^2$. Every term is nonincreasing in λ and tends to zero,

and a term with $c_j \neq 0$ decreases strictly. Some $c_j \neq 0$ iff $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$. As $\lambda \rightarrow 0$, $d_j/(d_j^2 + \lambda) \rightarrow 1/d_j$, and $\sum_{j \leq r} (c_j/d_j) \mathbf{v}_j = \mathbf{X}^+ \mathbf{y}$ by Theorem 10.4.1(b).

- (e) With σ^2 known, the posterior density is proportional to $\exp\{-(\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2)/(2\sigma^2)\}$. Completing the square as in the proof of Theorem 7.6.2, with $\mathbf{P}_0 = \lambda\mathbf{I}$ and $\mathbf{m}_0 = \mathbf{0}$,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2 = (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_\lambda^R)^\top (\mathbf{X}^\top \mathbf{X} + \lambda\mathbf{I})(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_\lambda^R) + C,$$

with C free of $\boldsymbol{\beta}$. This is the stated normal density. The last statement is Proposition 7.6.3(d). ■

Part (a) says what ridge does. It keeps the directions $\mathbf{v}_j$ and shrinks the coordinate along each by $f_j(\lambda)$: directions with $d_j^2 \gg \lambda$ are hardly touched, and those with $d_j^2 \ll \lambda$ are shrunk almost to zero, whatever the size of the coefficients. Comparing with the oracle factors $f_j^* = d_j^2/(d_j^2 + \sigma^2/\alpha_j^2)$ of Proposition 27.1.2(c), ridge is the oracle when all canonical coefficients have the same size, $\alpha_j^2 = \sigma^2/\lambda$. Part (e) says the same in Bayesian terms: the prior $N(\mathbf{0}, (\sigma^2/\lambda)\mathbf{I})$ treats every direction alike.

The effective degrees of freedom in (b) count each dimension by the fraction $f_j(\lambda)$ in which it is used; Chapter 29 uses them in model-selection criteria, and Exercise (C1) gives them a second meaning.

Remark (Penalized and constrained forms). For any penalty $P \geq 0$, a minimizer $\hat{\mathbf{b}}$ of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda P(\mathbf{b})$, with $\lambda > 0$, also minimizes $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $P(\mathbf{b}) \leq t$, where $t = P(\hat{\mathbf{b}})$. Indeed, if $P(\mathbf{b}) \leq t$, then

$$\begin{aligned} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 &\geq \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\{P(\mathbf{b}) - t\} \\ &\geq \|\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}\|^2 + \lambda\{P(\hat{\mathbf{b}}) - t\} = \|\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}\|^2. \end{aligned}$$

So ridge is least squares restricted to a ball, $\|\mathbf{b}\|^2 \leq t$, whose radius is fixed by λ (Exercise (B3)). The lasso of Section 27.4 replaces the ball by a cross-polytope.

27.2.3 Ridge beats least squares

Exercise (B1) showed that the total mean squared error of ridge falls as λ leaves 0. Hoerl and Kennard made the range explicit, and Theobald (1974) added the matrix sense.

Theorem 27.2.3. Hoerl–Kennard

Let $\mathbf{X}$ have full column rank, $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$, $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ with $\sigma^2 > 0$, and let $\alpha_{\max}^2 = \max_j \alpha_j^2$ with $\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}$. Let $m(\lambda)$ be the total mean squared error Eq. 27.2.2, so that $m(0) = \sigma^2 \sum_j d_j^{-2}$ is that of least squares.

- (a) $m(\lambda) < m(0)$ for every $0 < \lambda < 2\sigma^2/\alpha_{\max}^2$, and in particular for every $0 < \lambda < 2\sigma^2/\|\boldsymbol{\beta}\|^2$. If $\boldsymbol{\beta} = \mathbf{0}$, for every $\lambda > 0$.
 (b) m is strictly decreasing on $[0, \sigma^2/\alpha_{\max}^2]$.

- (c) $\hat{\boldsymbol{\beta}}_\lambda^R$ is at least as good as $\hat{\boldsymbol{\beta}}$ in the matrix sense iff

$$\frac{\lambda}{\sigma^2} \sum_j \frac{d_j^2 \alpha_j^2}{2d_j^2 + \lambda} \leq 1,$$

which holds whenever $0 < \lambda \leq 2\sigma^2/\|\boldsymbol{\beta}\|^2$.

Proof.

- (a) Compare term by term. The j th term of Eq. 27.2.2 is below its value σ^2/d_j^2 at $\lambda = 0$ iff $d_j^2(\sigma^2 d_j^2 + \lambda^2 \alpha_j^2) < \sigma^2(d_j^2 + \lambda)^2$, that is, iff $\lambda^2 d_j^2 \alpha_j^2 < 2\lambda \sigma^2 d_j^2 + \lambda^2 \sigma^2$, or, dividing by $\lambda > 0$,

$$\lambda(d_j^2 \alpha_j^2 - \sigma^2) < 2\sigma^2 d_j^2.$$

If $d_j^2 \alpha_j^2 \leq \sigma^2$, this holds for every $\lambda > 0$. Otherwise it holds for $\lambda < 2\sigma^2 d_j^2/(d_j^2 \alpha_j^2 - \sigma^2)$, and the right side exceeds $2\sigma^2 d_j^2/(d_j^2 \alpha_j^2) = 2\sigma^2/\alpha_j^2 \geq 2\sigma^2/\alpha_{\max}^2$. So for $0 < \lambda < 2\sigma^2/\alpha_{\max}^2$ every term decreases, and so does the sum. Finally $\alpha_{\max}^2 \leq \|\boldsymbol{\alpha}\|^2 = \|\boldsymbol{\beta}\|^2$.

- (b) By Exercise (B1), in the present notation, $m'(\lambda) = 2 \sum_j d_j^2 (\lambda \alpha_j^2 - \sigma^2)/(d_j^2 + \lambda)^3$. For $0 \leq \lambda < \sigma^2/\alpha_{\max}^2$ every numerator is negative, so $m' < 0$ there, and m is strictly decreasing on the closed interval.
- (c) Write $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$ and $\mathbf{A} = (\mathbf{S} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$. In the notation of Theorem 27.1.1(c), all matrices involved have the eigenvectors $\mathbf{v}_j$, with eigenvalues

$$\Delta_{\mathbf{A}} : \sigma^2 \left(\frac{1}{d_j^2} - \frac{d_j^2}{(d_j^2 + \lambda)^2} \right) = \sigma^2 \frac{\lambda(2d_j^2 + \lambda)}{d_j^2(d_j^2 + \lambda)^2} > 0,$$

and the bias is $\boldsymbol{\delta} = -\sum_j \lambda \alpha_j (d_j^2 + \lambda)^{-1} \mathbf{v}_j$. So $\Delta_{\mathbf{A}}$ is positive definite and $\boldsymbol{\delta}^\top \Delta_{\mathbf{A}}^{-1} \boldsymbol{\delta} = \sum_j \lambda^2 \alpha_j^2 (d_j^2 + \lambda)^{-2} \cdot d_j^2 (d_j^2 + \lambda)^2 / \{\sigma^2 \lambda (2d_j^2 + \lambda)\}$, which is the stated sum. Each factor $d_j^2/(2d_j^2 + \lambda)$ is below $1/2$, so the sum is at most $\lambda \|\boldsymbol{\alpha}\|^2/(2\sigma^2) = \lambda \|\boldsymbol{\beta}\|^2/(2\sigma^2)$. Apply Theorem 27.1.1(c). ■

The theorem is an existence result. The good range of λ depends on the unknown $\boldsymbol{\beta}$ and σ^2 , and it says nothing about a λ chosen from the data. Replacing $\boldsymbol{\beta}$ by $\hat{\boldsymbol{\beta}}$ in the bound is misleading, because $\|\hat{\boldsymbol{\beta}}\|^2$ overestimates $\|\boldsymbol{\beta}\|^2$ by $\sigma^2 \sum_j d_j^{-2}$ on average (Exercise (B3)), which is large exactly when ridge is needed. What the theorem does establish is that least squares is never the best member of the ridge family.

27.2.4 Choosing the ridge parameter

Hoerl and Kennard suggested choosing the smallest λ at which the ridge trace “stabilizes”, a subjective judgement. Hoerl, Kennard and Baldwin (1975) proposed the plug-in $\lambda = p\hat{\sigma}^2/\|\hat{\boldsymbol{\beta}}\|^2$, the optimum of Exercise (A2) with estimates substituted; since $\|\hat{\boldsymbol{\beta}}\|^2$ is too large on average, it tends to shrink too little under collinearity. A more defensible choice minimizes an estimate of prediction error. Because ridge is a linear smoother, $\hat{\mathbf{y}} = \mathbf{H}_\lambda \mathbf{y}$, the leave-one-out prediction errors need only one fit: the residual of case i divided by $1 - h_{ii}(\lambda)$, with $h_{ii}(\lambda)$ the i th diagonal entry of $\mathbf{H}_\lambda$ (Exercise (B1); add $1/n$ for an unpenalized intercept), the ridge version of Theorem 20.3.1.

Generalized cross-validation replaces each $h_{ii}(\lambda)$ by the average $\text{tr } \mathbf{H}_\lambda/n$:

$$\text{GCV}(\lambda) = \frac{n^{-1} \|\mathbf{y} - \mathbf{H}_\lambda \mathbf{y}\|^2}{(1 - \text{tr } \mathbf{H}_\lambda/n)^2}$$

(with an unpenalized intercept, $\text{tr } \mathbf{H}_\lambda$ becomes $1 + \text{df}(\lambda)$). [Chapter 29](#) develops both criteria for general linear smoothers (Theorem 29.3.2 and Definition 29.3.2). A Bayesian alternative estimates the prior variance σ^2/λ from the marginal likelihood, an empirical Bayes idea that returns in [Section 27.5](#).

Example. Ridge regression for Longley's data

Longley's regression (Example (Longley's regression)) relates total employment, in thousands, to six economic series over $n = 16$ years. With the regressors standardized, the eigenvalues d_j^2 of $\mathbf{X}^\top \mathbf{X}$ run from 69.05 down to 0.0057. The least squares coefficients, in thousands of persons per standard deviation, are 163 for the GNP deflator, -3560 for GNP, -1888 for unemployment, -719 for the armed forces, -355 for population and 8709 for the year. The GNP and YEAR coefficients are large and of opposite sign: the two series rose together, and the data can hardly separate them.

[Figure 27.2.1\(a\)](#) shows the ridge trace against the effective degrees of freedom. As λ grows, the GNP and YEAR coefficients approach each other, and the GNP coefficient changes sign at $\lambda = 0.0110$, where $\text{df} = 5.1$. Leave-one-out cross-validation chooses $\lambda = 0.0026$, with $\text{df} = 5.61$, and GCV chooses $\lambda = 0.0030$, with $\text{df} = 5.56$. Both prefer little shrinkage. At the leave-one-out choice the coefficients are -6 , -1809 , -1645 , -669 , -825 and 7415 . The largest changes are in the coefficients of GNP, YEAR and population, the three series that rose together, and the deflator coefficient is essentially zero. The minimized root mean squared leave-one-out error is 402, against 425 for least squares, but this minimum is optimistic, since the same errors chose λ . Choosing λ afresh within each leave-one-out fold (*nested* cross-validation) gives an honest 441: on these sixteen years tuned ridge does not predict better than least squares.

Panel (b) asks what the theorem guarantees in a world where the truth is the least squares fit, $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$, with $\sigma = s = 304.9$. There, least squares has total mean squared error 19.33 million, and the best ridge estimate, at $\lambda = 0.0019$ with $\text{df} = 5.69$, has 15.01 million, a ratio of 0.777. [Theorem 27.2.3\(a\)](#) guarantees an improvement only for $\lambda < 0.00383$, and part (c) gives dominance in the matrix sense for $\lambda \leq 0.00200$; in fact the improvement continues up to $\lambda = 0.0058$. The ranges are narrow because this “truth” inherits the least squares estimate's large components along the weak directions.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = ["GNPDEFL", "GNP", "UNEMP", "ARMED", "POP", "YEAR"]
Z = data[names].to_numpy()
y = data["TOTEMP"].to_numpy()
n, p = Z.shape

X = (Z - Z.mean(axis=0)) / Z.std(axis=0, ddof=1)      # centred, unit standard
              deviation
yc = y - y.mean()                                     # centred response (the
              intercept is ybar)

U, d, Vt = np.linalg.svd(X, full_matrices=False)      # X = U diag(d) V^T
```

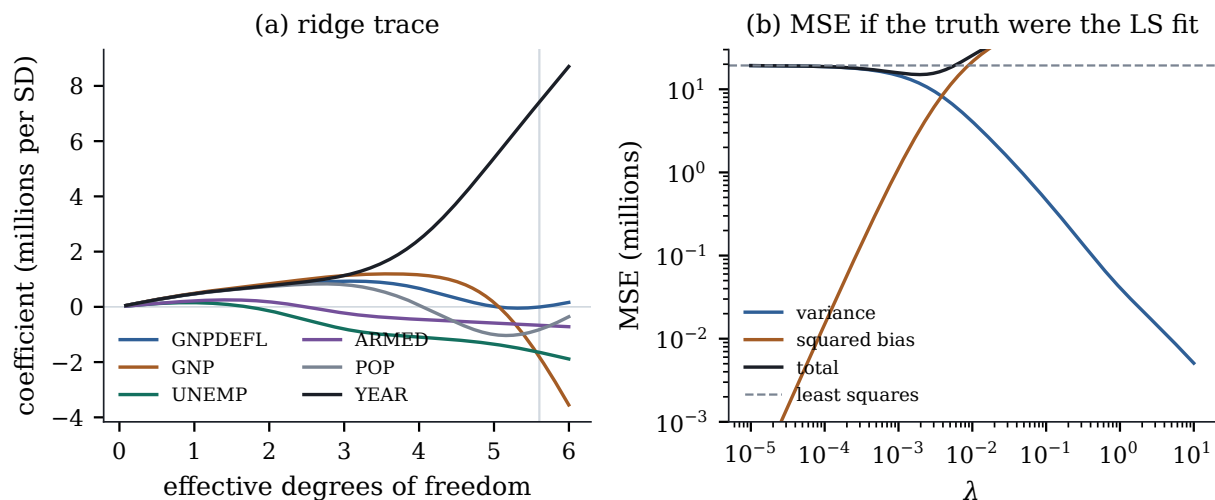


Figure 27.2.1: Ridge regression for Longley's data. (a) The ridge trace: standardized coefficients against the effective degrees of freedom $df(\lambda)$; the vertical line marks the leave-one-out choice. (b) Variance, squared bias and total mean squared error of ridge, computed exactly from Eq. 27.2.2 when the true coefficients are the least squares estimates and $\sigma = s$; the dashed line is least squares.

```
V = Vt.T
c = U.T @ yc                                     # coordinates of y along u_1,
..., u_p

def ridge(lam):
    """Ridge coefficients for the centred problem, through the SVD."""
    return V @ (d / (d ** 2 + lam) * c)

def edf(lam):
    """Effective degrees of freedom of the slopes: sum of the shrinkage factors."""
    return np.sum(d ** 2 / (d ** 2 + lam))

beta_ls = V @ (c / d)
print("singular values squared:", np.round(d ** 2, 4))
print("least squares:", np.round(beta_ls, 1))
for lam in [0.01, 0.1, 1.0]:
    print(f"lambda = {lam:5.2f}  edf = {edf(lam):.2f}  ridge =", np.round(ridge(lam), 1))
```

[Run this code online](#)

The second cell computes the leverages of the ridge smoother, the leave-one-out and GCV criteria, and the two choices of λ .

```
def hat_diag(lam):
    """Leverages of the ridge smoother with an unpenalized intercept."""
    f = d ** 2 / (d ** 2 + lam)
    return 1 / n + np.sum(U ** 2 * f, axis=1)

def loo_and_gcv(lam):
    resid = yc - X @ ridge(lam)
    h = hat_diag(lam)
    loo = np.mean((resid / (1 - h)) ** 2)                # leave-one-out shortcut
    gcv = np.mean(resid ** 2) / (1 - (1 + edf(lam)) / n) ** 2
```

```

return loo, gcv

grid = np.logspace(-4, 2, 601)
scores = np.array([loo_and_gcv(l) for l in grid])
lam_loo = grid[np.argmin(scores[:, 0])]
lam_gcv = grid[np.argmin(scores[:, 1])]
print(f"L00 chooses lambda = {lam_loo:.4f} (edf {edf(lam_loo):.2f})")
print(f"GCV chooses lambda = {lam_gcv:.4f} (edf {edf(lam_gcv):.2f})")
print("ridge at the L00 choice:", np.round(ridge(lam_loo), 1))

```

[Run this code online](#)

The script also checks the closed form against a direct solve, augmented least squares ([Exercise \(A1\)](#)) and the posterior mean of Theorem 7.6.2, and the leave-one-out shortcut against sixteen refits.

27.2.5 Exercises

27.2.5.1 A. Check your understanding

Exercise. A1

Show that $\hat{\beta}_\lambda^R$ is the least squares estimate for the augmented data $\tilde{\mathbf{y}} = (\mathbf{y}^\top, \mathbf{0}^\top)^\top$ and $\tilde{\mathbf{X}} = (\mathbf{X}^\top, \sqrt{\lambda}\mathbf{I})^\top$. Show that $\kappa(\tilde{\mathbf{X}})^2 = (d_1^2 + \lambda)/(d_p^2 + \lambda)$, so ridge can be computed stably by QR ([Chapter 10](#)), and that the augmented problem has “residual sum of squares” $\|\mathbf{y} - \mathbf{X}\hat{\beta}_\lambda^R\|^2 + \lambda\|\hat{\beta}_\lambda^R\|^2$.

Solution. $\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \|\mathbf{0} - \sqrt{\lambda}\mathbf{b}\|^2$, which is [Eq. 27.2.1](#), so the minimizers coincide and the minimum is the stated sum. $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$ has eigenvalues $d_j^2 + \lambda$, and the singular values of $\tilde{\mathbf{X}}$ are their square roots (Theorem 1.8.1). The condition number is the ratio of the largest to the smallest (Definition 1.8.1).

Exercise. A2

Suppose $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$. Show that $\hat{\beta}_\lambda^R = \hat{\beta}/(1 + \lambda)$, $\text{df}(\lambda) = p/(1 + \lambda)$, and that the value of λ minimizing the total mean squared error is $p\sigma^2/\|\beta\|^2$.

27.2.5.2 B. Practice

Exercise. B1

Let $\hat{\beta}_{\lambda,(i)}^R$ be the ridge estimate computed without case i (no intercept), and h_{ii} the i th diagonal entry of $\mathbf{H}_\lambda$. Show that

$$y_i - \mathbf{x}_{(i)}^\top \hat{\beta}_{\lambda,(i)}^R = \frac{y_i - \mathbf{x}_{(i)}^\top \hat{\beta}_\lambda^R}{1 - h_{ii}}.$$

Solution. Put $\mathbf{K} = \mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ and $\mathbf{x} = \mathbf{x}_{(i)}$, so $h_{ii} = \mathbf{x}^\top \mathbf{K}^{-1} \mathbf{x} < 1$. Without case i the matrix is $\mathbf{K} - \mathbf{x}\mathbf{x}^\top$ and the right side is $\mathbf{X}^\top \mathbf{y} - \mathbf{x}y_i$. By Eq. 1.6.7, $(\mathbf{K} - \mathbf{x}\mathbf{x}^\top)^{-1} = \mathbf{K}^{-1} + \mathbf{K}^{-1} \mathbf{x}\mathbf{x}^\top \mathbf{K}^{-1} / (1 - h_{ii})$. Hence

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}}_{\lambda, (i)}^R = \left(\mathbf{x}^\top \mathbf{K}^{-1} + \frac{h_{ii} \mathbf{x}^\top \mathbf{K}^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}y_i) = \frac{\mathbf{x}^\top \hat{\boldsymbol{\beta}}_\lambda^R - h_{ii} y_i}{1 - h_{ii}},$$

and y_i minus this is $(y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}_\lambda^R) / (1 - h_{ii})$. The argument is that of Theorem 20.3.1 with $\mathbf{X}^\top \mathbf{X}$ replaced by $\mathbf{K}$; only the rank-one update is used.

Exercise. B2

Generalized ridge regression uses a separate constant for each principal direction: $\hat{\boldsymbol{\beta}}_{\mathbf{K}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{V}\mathbf{K}\mathbf{V}^\top)^{-1} \mathbf{X}^\top \mathbf{y}$ with $\mathbf{K} = \text{diag}(k_1, \dots, k_p)$, $k_j \geq 0$. Show that its shrinkage factors are $d_j^2 / (d_j^2 + k_j)$, and that the choice $k_j = \sigma^2 / \alpha_j^2$ gives the oracle factors of Proposition 27.1.2(c).

Exercise. B3

Show that for every t with $0 < t < \|\hat{\boldsymbol{\beta}}\|^2$ (full column rank) there is exactly one $\lambda > 0$ with $\|\hat{\boldsymbol{\beta}}_\lambda^R\|^2 = t$, and that $\hat{\boldsymbol{\beta}}_\lambda^R$ is then the unique minimizer of $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ subject to $\|\mathbf{b}\|^2 \leq t$.

27.2.5.3 C. Going deeper

Exercise. C1

For any estimator $\hat{\boldsymbol{\mu}} = \mathbf{S}\mathbf{Y}$ of $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$ with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$, show that $\sigma^{-2} \sum_i \text{Cov}(\hat{\mu}_i, Y_i) = \text{tr } \mathbf{S}$. Deduce that $\text{df}(\lambda)$ measures how strongly the ridge fit follows its own data, and that $\mathbf{E}\|\mathbf{Y} - \mathbf{H}_\lambda \mathbf{Y}\|^2 = \mathbf{E}\|\mathbf{H}_\lambda \mathbf{Y} - \boldsymbol{\mu}\|^2 + n\sigma^2 - 2\sigma^2 \text{df}(\lambda)$.

Exercise. C2

Let $p > n$. Show that $\hat{\boldsymbol{\beta}}_\lambda^R = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$, which needs only an $n \times n$ inverse, and that $\hat{\boldsymbol{\beta}}_\lambda^R \in \mathcal{C}(\mathbf{X}^\top)$. What is the limit as $\lambda \rightarrow 0$? [Chapter 28](#) studies this limit.

Solution. The identity $\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n) = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top$ gives $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$ after multiplying by the two inverses. The right side is in $\mathcal{C}(\mathbf{X}^\top)$. By [Theorem 27.2.2\(d\)](#) the limit is $\mathbf{X}^+ \mathbf{y}$, the least squares solution of smallest norm, which interpolates the data when $\text{rank}(\mathbf{X}) = n$.